

JI-HOON OH (오지훈)

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Professional Experience

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| Dec. 2024 – Present | Postdoctoral Researcher
Scripps Institution of Oceanography, UC San Diego
Advisor: Prof. Shang-Ping Xie |
| Mar. 2024 – Nov. 2024 | Postdoctoral Researcher
School of Earth and Environmental Sciences, Seoul National University
Advisor: Prof. Jong-Seong Kug |

Education

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|-----------------------|---|
| Mar. 2021 – Feb. 2024 | Ph.D.
Department of Environment Science and Engineering, POSTECH
Advisor: Prof. Jong-Seong Kug |
| Mar. 2019 – Feb. 2021 | M.S.
Department of Environment Science and Engineering, POSTECH
Advisor: Prof. Jong-Seong Kug |
| Mar. 2013 – Feb. 2019 | B.S.
Atmospheric Science, Pusan National University
<i>* Mandatory Military Service (Mar. 2014 – Feb. 2016)</i>
<i>Republic of Korea Navy, ROKS Seongnam Patrol Combat Corvette 775</i> |

Publications

* : papers where I am the (co-)corresponding author

† : equally contributed

In Progress

5. Q. Ren, S.-P. Xie, A. Hu, **J.-H. Oh**, Z.-H. Fu, Y. Li, F. Wang: An intermediate state of Atlantic Meridional Overturning Circulation anchored by warm subtropical deep convection (*submitted*)
4. Y. Shin, N. Boers, Y. Huang, J. Ko, **J.-H. Oh**, J.-S. Kug: AI emulator highlights underestimated risk of AMOC collapse in current climate models (*Science Advances*; *under review*)
3. **J.-H. Oh**, S.-P. Xie, J.-S. Kug, J.-H. Park, Y. Yamagami, A. Miyamoto, G.-I. Kim, B. Wu: Atlantic Meridional Overturning Circulation slowdown amplifies decadal variability along the Kuroshio-Oyashio and Gulf Stream extensions (*Science Advances*; *in revision*)
2. **J.-H. Oh***, and S.-P. Xie: Centennial southward migration of tropical rain belt in a post-zero emissions world (*Communications Earth & Environment*; *in revision*)

1. J.-S. Kug, Y. Shin, **J.-H. Oh**, C. Liu, S.-S. Son, S.-W. Yeh, S.-I. An, S.-K. Min, J. Schwinger, H. Lee, N.J. Steinert, K. Zickfeld: Climate irreversibility and transient hysteresis under CO₂ removal (*Nature Climate Change*; in revision; ***review paper**)

Peer reviewed

22. T. Iwakiri, J.-S. Kug, S.-I. An, Y. Shin, **J.-H. Oh**, S. Wang (2026): AMOC slowdown amplifies North Atlantic salinity variability to unprecedented levels, *Nature Communications*, 17, 7021 [\[link\]](#)
21. S. Wang, Y. Shin, **J.-H. Oh**, H. Kim, J.-S. Kug (2026): Critical role of low cloud feedback in irreversible sea level rise, *Nature Communications*, 17, 6292 [\[link\]](#)
20. D. Lee, **J.-H. Oh**, J. Kam, S.-W. Park, J.-S. Kug (2026): Biophysical impacts of urbanization on climate change and vegetation in Borneo island, *Journal of Environmental Management*, 398, 128446 [\[link\]](#)
19. **J.-H. Oh**, J.-S. Kug, Y. Shin, X. Geng, S. Wang, F.-F. Jin, S.-I. An, S.-P. Xie, W. Liu (2025): Noise-induced tipping of Atlantic Meridional Overturning Circulation under climate mitigation scenarios, *Nature Communications*, 16, 11515 [\[link\]](#) [\[News 🇰🇷\]](#) [\[News 🇺🇸\]](#) [\[Video 🇰🇷\]](#)
18. X. Geng, **J.-H. Oh**, Y. Shin, J. Shin, K.-M. Noh, J.-S. Kug, S.-W. Park (2025): Non-monotonic future changes in the North Atlantic warming hole under a fast CO₂ emission scenario, *Journal of Climate*, 38, 21, 6011-6023 [\[link\]](#)
17. J.-H. Park, J.-H. Koo, J.-S. Kug, S.-J. Lee, M.-K. Sung, J. Kim, E.-C. Chang, Y.-M. Yang, S.-S. Park, K.-H. Kwak, **J.-H. Oh**, H.-J. Kang, S.-I. An (2025): Arctic stratospheric ozone as a precursor of ENSO events since 2000s, *npj Climate and Atmospheric Science*, 8, 1, 338 [\[link\]](#)
16. J. Piao, W. Chen, J.-S. Kug, S. Chen, **J.-H. Oh**, L. Wang, Q. Cai (2025): Land aridification persists in vulnerable drylands under climate mitigation scenarios, *Communications Earth & Environment*, 6, 1, 732 [\[link\]](#)
15. D.-G. Lee, **J.-H. Oh**, J. Kam, J.-S. Kug (2025): Phytoplankton blooming mechanisms over the East China Sea during post El-Niño summers, *Biogeosciences*, 22, 13 [\[link\]](#)
14. H. Lee, K.-M. Noh, **J.-H. Oh**, S.-W. Park, Y. Shin, J.-S. Kug (2025): Emergence of an oceanic CO₂ uptake hole under global warming, *Nature Communications*, 16, 3199 [\[link\]](#)
13. H.-J. Kim, J.-S. Kim, S.-I. An, J. Shin, **J.-H. Oh**, J.-S. Kug (2024): Pervasive fire danger continued under a negative emission scenario, *Nature Communications*, 15, 10010 [\[link\]](#)
12. S. Wang, Y. Shin, **J.-H. Oh***, J.-S. Kug* (2024): Fast recovery of North Atlantic sea level in response to atmospheric CO₂ removal, *Communications Earth & Environment*, 5, 652 [\[link\]](#)
11. D.-G. Lee, **J.-H. Oh***, J.-S. Kug* (2024): Delayed ENSO impact on phytoplankton variability over the Western-North Pacific Ocean, *Environmental Research Communications*, 6, 10, 101006 [\[link\]](#)
10. G.-I. Kim, **J.-H. Oh**, N.-Y. Shin, S.-I. An, S.-W. Yeh, J. Shin, J.-S. Kug (2024): Deep ocean warming-induced El Niño changes, *Nature Communications*, 15, 1, 6225 [\[link\]](#)
9. Y. Shin, X. Geng, **J.-H. Oh**, K.-M. Noh, E. K. Jin, J.-S. Kug (2024): Fast and slow responses of Atlantic Meridional Overturning Circulation to Antarctic meltwater forcing, *Geophysical Research Letters*, 51, 9, e2024GL108272 [\[link\]](#)
8. **J.-H. Oh**, J.-S. Kug, S.-I. An, F.-F. Jin, M. McPhaden, J. Shin (2024): Emergent climate change patterns originating from deep ocean warming in climate mitigation scenarios, *Nature Climate Change*, 14, 3, 260-266 [\[link\]](#) [\[Video1 🇰🇷\]](#) [\[Video2 🇰🇷\]](#) [\[News 🇰🇷\]](#)
7. K.-M. Noh, **J.-H. Oh**, H.-G. Lim, J.-S. Kug (2024): Role of Atlantification in enhanced primary productivity in

the Barents Sea, *Earth's Future*, 12, 1, e2023EF003709 [\[link\]](#)

6. Pathirana, G, **J.-H. Oh**, W. Cai, S.-I. An, S.-K. Min, S.-Y. Jo, J. Shin, J.-S. Kug (2023): Increase in convective Extreme El Nino events in the CO₂ removal scenario, *Science Advances*, 9, 25, eadh2412 [\[link\]](#)
5. D.-G. Lee, **J.-H. Oh**, K.-M. Noh, E.-Y. Kwon, Y.-H. Kim, J.-S. Kug (2023): What controls the future phytoplankton change over Yellow and East China Seas under global warming?, *Frontiers in Marine Science*, 10, 1010341 [\[link\]](#)
4. **J.-H. Oh**, S.-I. An, J. Shin, J.-S. Kug (2022): Centennial memory of the Arctic Ocean for future Arctic climate recovery in response to carbon dioxide removal, *Earth's Future*, 10, 8, e2022EF002804 [\[link\]](#)
3. **J.-H. Oh**, K.-M Noh, H.-G Lim, E. K. Jin, S.-Y. Jun, J.-S. Kug (2022): Antarctic meltwater-induced dynamical changes in phytoplankton in the Southern Ocean, *Environmental Research Letters*, 17, 2, 024022 [\[link\]](#)
2. J.-S. Kug[†], **J.-H. Oh**[†], S.-I. An, S.-W. Yeh, S.-K. Min, S.-W. Son, J. Kam, Y.-G. Ham, J. Shin (2022): Hysteresis of intertropical convergence zone to CO₂ forcing, *Nature Climate Change*, 12, 1, 47-53 [\[link\]](#) [\[News 🌐\]](#) [\[News in NCC\]](#)
1. **J.-H. Oh**, W.-S. Park, H.-G. Lim, K.-M. Noh, E. K. Jin, J.-S. Kug (2020): Impact of Antarctic meltwater forcing on East Asian climate under greenhouse warming, *Geophysical Research Letters*, 47, 21, e2020GL089951 [\[link\]](#) [\[News 🌐\]](#)

Selected Presentations

- Jul. Centennial southward migration of the tropical rain belt in a post zero-emissions world, **UCLA**
2026: **Atmospheric & Oceanic Sciences seminar** (Los Angeles, USA; **Invited*)
- Oct. Irreversibility of global climate system and its potential predictability, **Young Scientist Special**
2024: **Session at KMS Fall Meeting** (Busan, South Korea)
- Apr. Forthcoming tipping point of the AMOC collapse despite climate mitigation efforts, **NORCE-SNU-**
2024: **KOPRI Joint Workshop** (Bergen, Norway)
- Feb. Deep ocean warming-induced irreversible climate changes, **Ocean Science Meeting** (New Orleans,
2024: USA)
- May Hysteresis of global climate system and its potential predictability, **KOPRI Wednesday Seminar**
2023: (Online, **Invited*)
- Mar. Deep ocean warming-induced irreversible climate changes, **POSTECH-Yonsei-UTokyo Joint**
2023: **Workshop** (Jeju, South Korea)
- Jan. Centennial memory of the Arctic Ocean for future Arctic climate recovery in response to carbon
2023: dioxide removal, **IRCC-KIST-IPRC Joint Workshop** (Hawaii, USA)
- Aug. Centennial memory of the Arctic Ocean for future Arctic climate recovery in response to carbon
2022: dioxide removal, **Model-Hierarchy Workshop** (San Jose, USA, Poster)
- Feb. Centennial memory of the Arctic Ocean for future Arctic climate recovery in response to carbon
2022: dioxide removal, **Ocean Science Meeting** (Online, Poster)
- Feb. Impact of Antarctic meltwater forcing on East Asian climate under greenhouse warming, **Ocean**
2020: **Science Meeting** (San Diego, USA, Poster)

Selected Awards and Grants

- Sep. 2025: **Grant (RS-2025-02653909) from National Research Foundation of Korea (NRF) Basic Science Research Program**, Principal Investigator, ~43,000 USD, Sep 2025 – Aug 2026
- Oct. 2024: **Best Doctoral Dissertation Award**, Korean Metrological Society (KMS)
- Feb. 2024: **Best Dissertation Award in the field of Natural Science** (2023 Sung-Kee Chung Award), Pohang University of Science and Technology (POSTECH)

Professional Activities and Service

- **Peer Reviews:** *Journal of Climate*, *Nature Climate Change*, *Geophysical Research Letters*, *Communications Earth & Environment*, *Nature Communications*, *Journal of Geophysical Research: Atmosphere*, *PLOS climate*